GLOBAL AI VISA ASSISTANT

End-to-End SaaS Product & AI Agent Build Blueprint

Implementation specification, developer handoff, workflows, APIs, database, privacy, testing, deployment, and a copy-ready AI coding-agent master prompt.

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Product concept: a multi-user, privacy-first visa research, manual checklist, optional document review, preparation, interview practice and voice-assistant platform.

IMPORTANT: This is a product specification, not legal or immigration advice. Rules, fees and eligibility must be checked against current official sources. No AI system should promise visa approval.

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1. Product vision and non-negotiable requirements

Build a responsive, production-oriented multi-tenant web SaaS. Each person owns an account and one or more visa applications. Users may complete the entire preparation flow WITHOUT uploading documents. Optional uploads unlock limited document-content review. AI must ground requirements in official sources and guide the next steps with uncertainty clearly shown.

• Personal accounts, secure login, separate data per user, optional consultant organizations with explicitly granted client access.

• Country, nationality, legal residence, consular jurisdiction, visa category, purpose, dates and applicant circumstances determine applicable rules.

• Manual checklist status: Not started / I have it / Need help / Not applicable (with reason) / Reviewed. Never equate self-reported possession with verification.

• Optional document uploads: explicit opt-in, private storage, delete/export, manual alternative, clear limitations.

• Official-source requirement research with provenance, update dates, citations, and human review before publishing changes.

• AI step-by-step visa preparation planner, truthful interview practice (text and voice), reminders and progress history.

• No autonomous visa submissions, payments, emails to third parties, or account changes without specific user authorization.

2. Personas and roles

Role	Permissions
Guest	View marketing pages and limited public overview; no private records.
Applicant	Own applications, checklist, AI chat, optional documents, reminders, voice practice.
Consultant	Manage explicitly authorized clients in assigned organization; request access and view audit history.
Organization admin	Manage team, subscription, permissions and organization settings.
Platform admin	Manage curated rule sources, review queue, plans, abuse and system health; private documents not accessible by default.

3. User journey / product screens

• Landing -> signup/login -> email verification -> consent and privacy preferences.

• Create application: nationality, current legal residence, destination, consulate/location, visa category, purpose, study/work profile, intended dates.

• System matches a verified ruleset or shows "requirements not verified" and requests review; never fabricates a checklist.

• Requirements screen: each item has title, explanation, applicability, official source URL, last verified date, action guidance.

• Manual checklist: check item as available, mark missing, add notes; upload is optional and never a gate.

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• Optional document review: consent -> upload -> malware check -> OCR -> AI extracts possible fields -> user reviews and corrects -> flags.

• Preparation roadmap: prioritized next actions, official forms and appointment links, reminders and deadlines.

• AI chat/voice: ask "What is missing?", "What next?", "Explain bank evidence"; answers cite source or say uncertain.

• Interview practice: question bank by category, user response, clarity and consistency feedback; no invented narratives.

• Account settings: subscriptions, export, deletion, consent revocation, language, notifications.

4. MVP scope vs later releases

Release	Included	Not included
MVP (6-10 weeks)	5 destinations, one visa category per jurisdiction, manual checklist, verified sources, personal dashboard, AI chat, roadmap, reminders, secure auth.	Uploads, consultant teams, live voice, browser automation.
V1.1	Optional document OCR/review, PDF summary, consultant workspace, subscription billing.	Automatic filing or eligibility guarantees.
V2	Voice interaction, multi-language UI, source-change alerts, more countries, analytics.	Unreviewed automatic legal-rule publication.

Suggested initial country set: Italy, Germany, UK, Canada and Australia, focused on study visas for residents of Pakistan; expand only after validating jurisdiction-specific rules. Long-term destination catalog may contain 30+ countries, but "listed" must not mean "fully verified".

5. Core functional modules

Identity & privacy

Email verification, password reset, session management, optional MFA, export/delete, per-feature consent.

Application profile

Versioned application answers; nationality and residence treated separately; multiple applications per account.

Requirements finder

Official source registry, fetch/parsing, country/visa/jurisdiction rule matching, citation links, human review queue.

Manual checklist

Statuses, notes, timestamps, source snapshots, item dependencies, progress based on self-report.

Optional AI document review

Encrypted uploads, virus scanning, OCR, field extraction, confidence flags, comparison, deletion.

AI guidance & roadmap

Retrieval-grounded Q&A, personalized tasks, deadlines, next-step explanations, refusal to guess.

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Visa interview coach

Truthful practice questions, speech/text answers, feedback on clarity, grammar and consistency.

Voice task assistant

Speech-to-text -> authenticated task planning -> safe read tools -> spoken summary; confirmation before writes.

Consultant portal

Client invitations, consent grants, case assignment, comments, access expiry and audit logs.

Billing & admin

Plans, quotas, payment webhook verification, support tickets, source review and system metrics.

6. Verified official sources pipeline

• Curate an allowlist of official immigration, embassy, consulate and authorized application-provider URLs. Official government/embassy sources take priority.

• Scheduled fetch via allowed APIs, RSS, HTML or downloadable PDFs; respect robots, terms, rate limits, CAPTCHA and access restrictions. Never bypass access controls.

• Store original URL, page title, jurisdiction, language, fetch timestamp, publication/effective date if provided, checksum, extracted text and snapshot reference.

• AI extracts candidate requirements into structured JSON with exact supporting text spans and URLs. Extraction is NOT approval.

• Rule validator checks missing citations, conflicting dates, country/visa applicability, stale sources and extraction confidence.

• Human reviewer approves material updates before they become public; maintain rule version history and rollback.

• If no trusted rule exists or the source is inaccessible, show "Not verified" and official link; avoid making up amounts, fees or eligibility.

• Cache approved rules. User requests query the curated database first; do not browse every official website on every user action.

Rule JSON contract (conceptual)

rule_id, destination_country, nationality_scope, residence_country, consular_jurisdiction, visa_type, applicant_conditions, document_name, required_when, instructions, source_url, source_excerpt, source_published_at, verified_at, valid_from, review_status, version, confidence_note.

7. AI agent tools and safety contracts

Tool	Behavior / constraint
search_official_sources	Search allowlisted sources; returns links and timestamps.
get_applicable_rules	Returns only approved matching rule versions; returns unknown when no match.

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Tool	Behavior / constraint
get_user_checklist	Read authorized user checklist; never read other tenants.
set_checklist_status	Write only after user instruction; log before/after.
analyze_uploaded_document	Only for consented uploads; show extraction uncertainty.
create_preparation_plan	Generate tasks using verified rules and user-selected statuses.
create_reminder	Only after user confirms date/time and permission.
draft_message	Prepare message, but do not send without approval.
run_mock_interview	Practice, assess language and clarity; avoid misleading coaching.

Tool calls must be server-side, schema validated, authenticated and authorization checked. Prompt-injection text in websites/PDFs is untrusted. Use bounded execution, timeouts, retries, budgets, audit logs and approval gates. Do not give the model unrestricted shell, filesystem, browser login or payment privileges.

8. Technical architecture

Frontend: Next.js App Router, React, TypeScript, Tailwind CSS, shadcn/ui. Backend: Next.js route handlers or a separate Node.js service; Zod validation. Data: PostgreSQL + Prisma with tenant isolation. Object storage: private S3-compatible buckets. Queue: Redis + BullMQ for scheduled source refresh/OCR/reminders. AI: configurable OpenAI/Gemini-compatible provider through a server-side adapter. OCR: provider adapter (Document AI / Textract / other). Email: transactional email provider. Voice: speech-to-text and text-to-speech APIs; optional realtime voice in V2. Monitoring: structured logs, error tracking, metrics.

Request flow: browser -> authenticated API -> application service -> approved rules database -> AI retrieval/plan service -> response with citations. Optional files: browser -> short-lived signed upload URL -> private storage -> malware scan -> OCR worker -> extraction review -> encrypted result. Background rules: scheduled fetch -> parse -> diff -> human review -> approved version -> alerts.

9. Database schema: minimum entities

Entity	Essential fields
users	id, email, password_hash/identity_id, locale, created_at, deleted_at
organizations	id, name, owner_user_id, plan_id
memberships	id, organization_id, user_id, role, status
applications	id, owner_id, organization_id?, destination, nationality, residence, jurisdiction, visa_type, status
source_documents	id, official_url, title, fetched_at, effective_at, checksum, content_ref
rulesets / requirements	id, scope fields, document_name, conditions_json, source_id, version, approval_status
checklist_items	id, application_id, requirement_id, status, notes, updated_at

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Entity	Essential fields
uploaded_documents	id, owner_id, application_id, object_key, consent_id, retention_until, status
document_reviews	id, document_id, extraction_json, flags_json, reviewed_by, created_at
ai_conversations	id, application_id, user_id, messages_ref, model, token_usage
tasks / reminders	id, application_id, title, due_at, status, channel
consents / audit_logs	id, actor_id, action, resource_id, timestamp, metadata
subscriptions / usage	id, organization_id/user_id, provider_id, plan, billing_status, quota

Use UUIDs, foreign keys, indexed scope fields, row-level security where appropriate, encryption at rest, short-lived signed URLs, soft deletion only where retention is justified, and hard deletion workflows for sensitive files.

10. REST API endpoints (illustrative)

Method / path	Purpose
POST /api/auth/register	Register account, verify email
POST /api/applications	Create application with validated profile
GET /api/applications/:id	Read own/authorized application
GET /api/applications/:id/requirements	Approved rules + provenance + verification state
PATCH /api/checklist/:id	Update own manual status and note
POST /api/ai/chat	Grounded guidance with citations and usage metering
POST /api/documents/upload-intent	Consent check, secure signed upload
POST /api/documents/:id/analyze	Queue malware scan and OCR review
DELETE /api/documents/:id	Delete file and related extracts per retention policy
POST /api/interview/session	Start practice session
POST /api/voice/transcribe	Speech to text with consent
POST /api/tasks	Create approved reminder/task
POST /api/billing/webhook	Verify signed payment event
GET /api/admin/rules/review	Restricted reviewer queue

Use rate limiting, request schemas, idempotency for writes, pagination, CSRF protection where needed, signed webhooks, and server-side authorization on every endpoint.

11. Suggested repository structure

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apps/web/app/(auth), (dashboard), (marketing), api

apps/web/components/checklist, requirements, chat, voice, interview

packages/db/prisma/schema.prisma and migrations

packages/core/application, checklist, permissions, billing

packages/ai/providers, prompts, retrieval, tool-registry

packages/sources/connectors, parsers, diff, review

workers/ocr, source-refresh, reminders

infra/docker, deployment, monitoring

tests/unit, integration, e2e, security

docs/architecture, api, privacy, source-governance

12. Build sequence: AI coding agent implementation plan

Phase 0 - Foundations

Initialize monorepo, lint, format, environment validation, CI, Docker Compose PostgreSQL/Redis, README, threat model.

Phase 1 - Identity

Auth, account/profile, session security, consent settings, access-control tests.

Phase 2 - Visa applications

Country and visa selection, jurisdiction questions, CRUD, versioned applicant profile.

Phase 3 - Sources and rules

Allowlist, source parser, structured extraction, reviewer approval, approved-rule query, stale warning.

Phase 4 - Manual checklist

Generate checklist from approved rules, status toggles, progress, no-upload flow, export.

Phase 5 - AI guide

Provider abstraction, grounded retrieval, cited answers, task planner, unknown handling.

Phase 6 - Optional documents

Private upload, malware scan, OCR, extraction review, retention/deletion.

Phase 7 - Preparation

Action plan, reminders, interview practice, PDF report, optional voice.

Phase 8 - SaaS business

Plans, quotas, billing adapter, consultant permissions, admin dashboard.

Phase 9 - Launch

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Automated tests, penetration review, accessibility, backup/restore, observability, pilot feedback.

Agent instruction: implement one phase at a time. After each phase, run tests, provide changed files, migration instructions, screenshots where possible, known limitations and a clear acceptance checklist. Never claim an external API works without credentials and sandbox verification.

13. API keys and external service setup

Service	You provide / setup
LLM provider	OpenAI or Gemini API key, model, budget cap, region/data policy
Search provider	Approved search API key or permitted source discovery approach
OCR	Document processing service credentials, billing, retention policy
Object storage	Private bucket, access keys, encryption, lifecycle rules
Email	Transactional provider API key, verified sender domain
Voice (later)	STT/TTS credentials, explicit mic permissions
Payments (later)	Merchant account, test/live credentials, signed webhook secret
Infrastructure	Postgres URL, Redis URL, domain, TLS, monitoring credentials

NEVER paste real API keys into AI prompts, repository files or client-side code. Set secrets in a secure environment manager, use .env.example with placeholders, restrict key scopes and rotate exposed keys. Payment provider availability varies by business location and onboarding eligibility.

14. Privacy, security and compliance acceptance gates

• No-upload path must be fully usable; manual checklist data stored without passport or financial document images.

• Separate consent for document processing, cloud AI analysis, voice recording and notifications; provide meaningful refusal alternatives.

• Tenant isolation tests: user A cannot access user B applications, documents, chats, signed URLs or reports.

• Private objects, encryption, malware scanning, content-type verification, size caps and safe PDF/image parsing.

• Delete/export tools; published privacy notice, retention periods and incident response procedure.

• Avoid claiming GDPR or other jurisdictional compliance without qualified review; assess data transfers and processors.

• AI output must distinguish "user says available", "AI observed", "official rule verified" and "not verified".

• No automatic visa application submission, payment, external messaging or irreversible actions without explicit approval.

• Audit administrative access and require reviewer approval for rule updates; backup and restore drills.

15. Test cases and definition of done

• A new applicant can sign up and create an Italy study visa case without uploading a single file.

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- A manual checkbox persists after logout/login and shows self-reported status, not verified status.
- Unverified source results never appear as authoritative mandatory requirements.
- Each verified requirement links to an official source and a verified timestamp.
- Changing nationality or residence re-evaluates the rule scope; unsupported combinations show unknown.
- A hostile instruction embedded in a PDF/website cannot trigger tool calls or data exfiltration.
- Cross-user access attempts return 403/404 and produce security logs.
- Optional uploaded file can be deleted; review and cached extracts follow retention/deletion policy.
- Voice assistant asks for confirmation before changing checklist, sending messages or scheduling reminders.
- Billing webhook rejects invalid signatures and handles duplicate events safely.
- Tests: unit, API integration, Playwright e2e, accessibility, performance and security.

16. Budget and operating assumptions

Self-built MVP planning range: PKR 25,000-60,000 upfront cash and PKR 5,000-20,000 monthly infrastructure/API spend at small scale, excluding your labor, formal security/legal reviews, acquisition, taxes and provider onboarding. A mature 30+ country platform can cost substantially more due to rule research and ongoing verification. These are illustrative estimates, not quotes. Start with one narrow visa route and pilot customers before expanding.

17. Copy-ready master prompt for your AI coding workflow

You are a senior full-stack engineer, SaaS architect, security engineer, and product QA lead. Build a production-oriented application named Global AI Visa Assistant based on this entire blueprint. Use Next.js App Router + TypeScript + Tailwind + shadcn/ui, Node server routes, PostgreSQL + Prisma, Redis/BullMQ for workers, private S3-compatible storage, and a configurable AI provider adapter (Gemini/OpenAI). Create real, runnable code and migrations; do not substitute mock UI for working backend flows. Prioritize a fully functional no-document-upload mode.

MANDATORY: each user has a private account and application(s); manual document checklist statuses persist; official visa requirements must be source-grounded, jurisdiction-specific, versioned and approved before public use; unknown requirements must be labeled unverified. Optional documents require explicit consent, secure storage and deletion. AI guides next steps and interview preparation with source citations and never guarantees approvals. Voice commands may read authorized data; write actions require confirmation.

DELIVER IN PHASES: 0 foundation; 1 auth; 2 applications; 3 official source registry/reviewer workflow; 4 manual checklist; 5 AI retrieval-guidance; 6 optional OCR; 7 interview/reminders/voice; 8 billing/consultant; 9 testing/deployment. Start with MVP phases 0-5. For each phase, give files changed, setup commands, tests, acceptance criteria and remaining risks. Do not claim integrations work until sandbox tested.

SECURITY: implement per-resource authorization, tenant isolation, validation, rate limits, encryption, signed URLs, malware scan, privacy consent, audit logs, prompt injection defense, secrets only in server environment, webhook signature verification, export/deletion, and safe error handling.

REQUIRED OUTPUTS: architecture.md, requirements.md, API specification, Prisma schema/migrations, .env.example (placeholders only), seed of DEMO rules clearly labeled NOT OFFICIAL, automated tests, Docker local setup, deployment guide, admin rule-review UI, applicant dashboard, source-cited AI responses, and release checklist. Ask for missing provider choices and credentials only when a phase actually requires them. Never request actual secrets in chat or commit them to git.

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SUCCESS TEST: new user creates an application, views approved source-linked requirements, checks documents manually without upload, gets a personalized next-step plan, can chat with AI about missing items, and sees a clear unknown/verification warning when authoritative rules are absent. All flows must work end-to-end with tested backend APIs.

18. Project handoff checklist

• Give AI agent this PDF and request implementation of phases 0-5 first.

• Choose initial visa route(s), supported languages, company name and branding.

• Decide hosting provider and AI provider; create accounts and secret environment variables yourself.

• Provide permitted official sources and reviewer process; verify legal and immigration content manually.

• Request a working demo URL, source repository, test report, and deployment instructions before accepting work.

• Pilot with 5-10 users; log confusion, incorrect rule matches, document privacy issues and support requests.

• Only expand destinations and paid subscriptions after accuracy, security and onboarding meet acceptance criteria.

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